

## Task 1 – Centralized Device Administration (TACACS+)

- Configure TACACS+ server in VLAN 200.
- Implement AAA on CS-SW1, CS-SW2, and Dist01.
- Create NetAdmin (priv 15) and HelpDesk (restricted) roles.
- Configure authentication, authorization, and accounting.
- Implement local fallback account.
- Validate login success, restriction enforcement, and TACACS failure behavior.

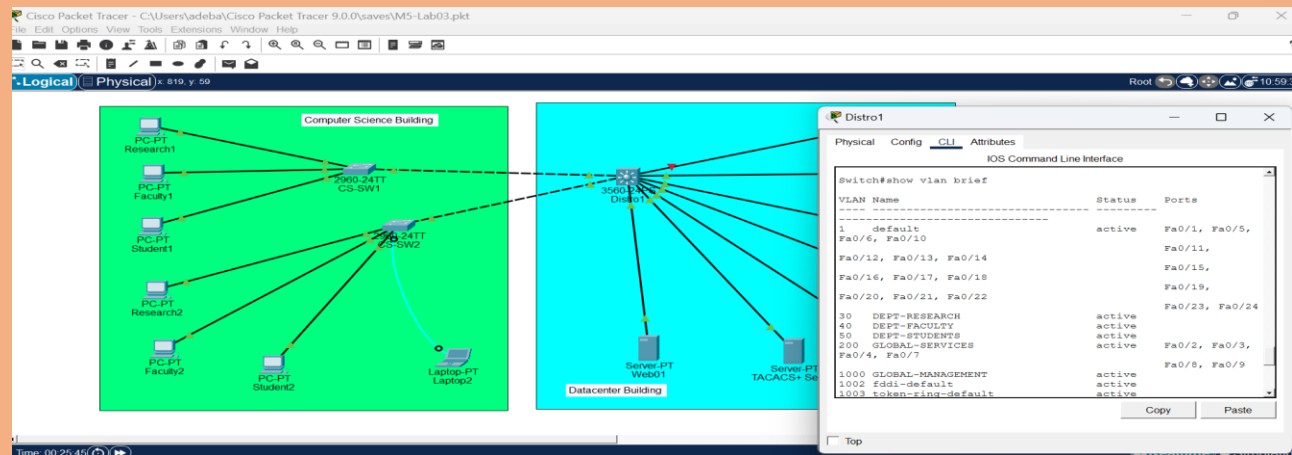


Figure 1: image showing the 3 new servers added to the vlan 200, with show vlan brief

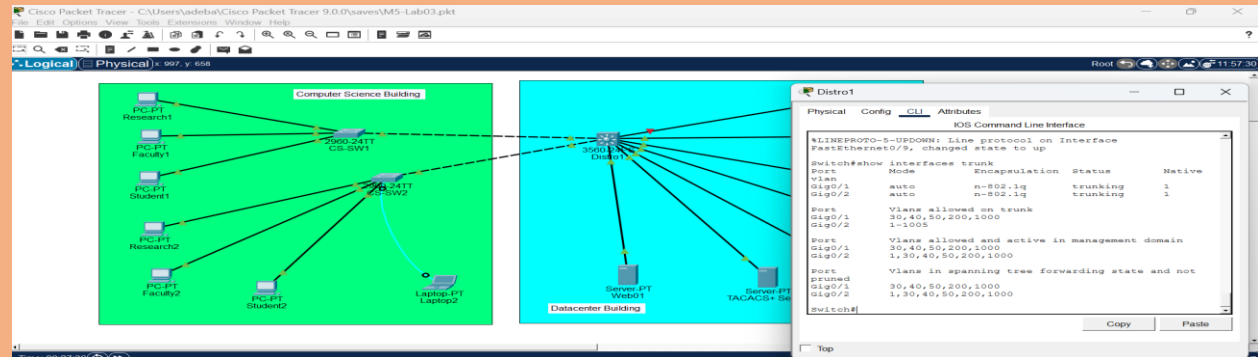


Figure 1b: Image show interface trunk after adding the 3 new servers

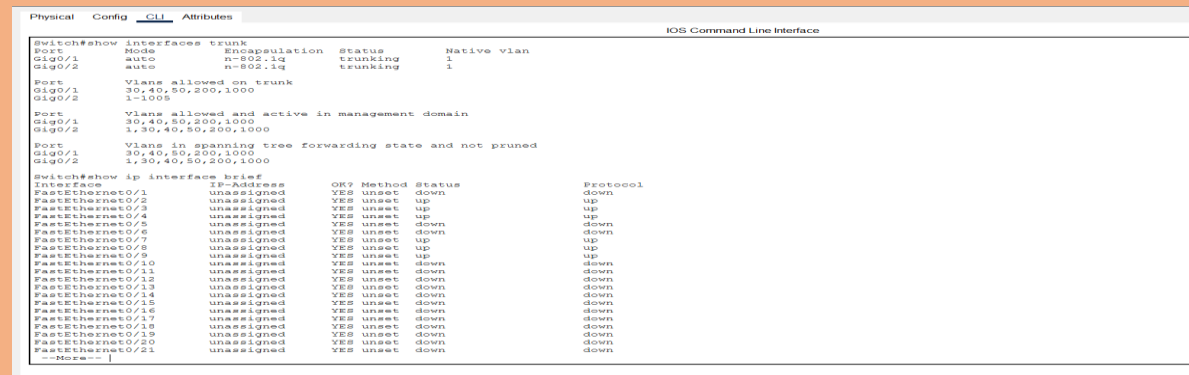


Figure 1c: image of show interface brief to show that all the servers are all up and running and the new servers are in port 7, 8 and 9 respectively

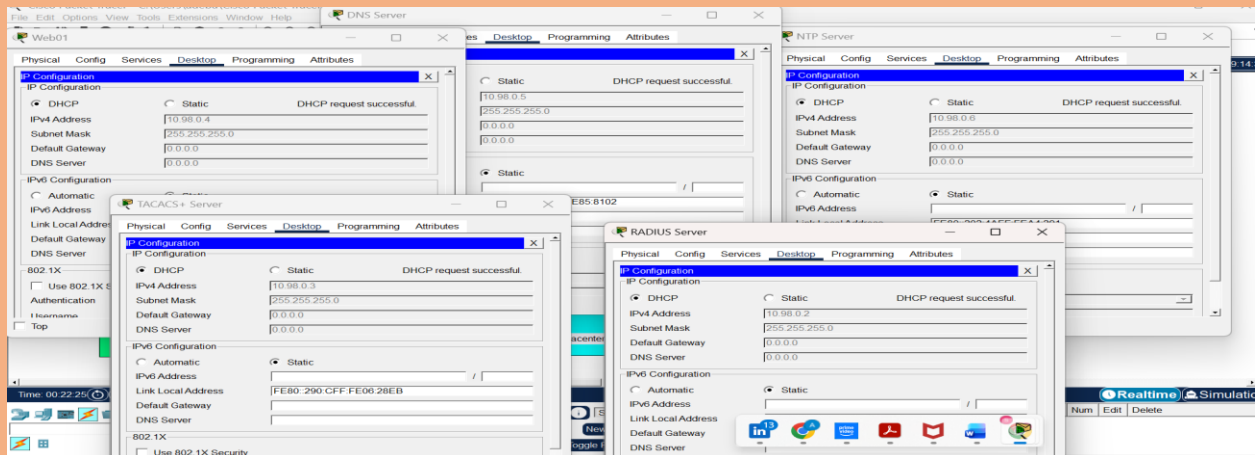


Figure 2a: showing all servers, including the new Servers are correctly receiving IP dynamically

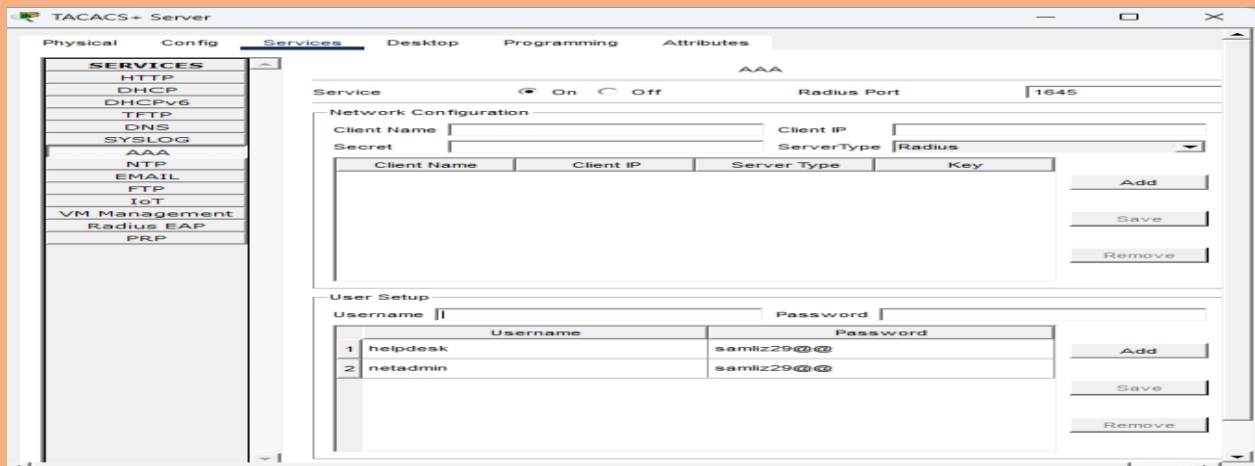


Figure 2b: Image showing the creation of the two user helpdesk and netadmin on the tacacs+ server.

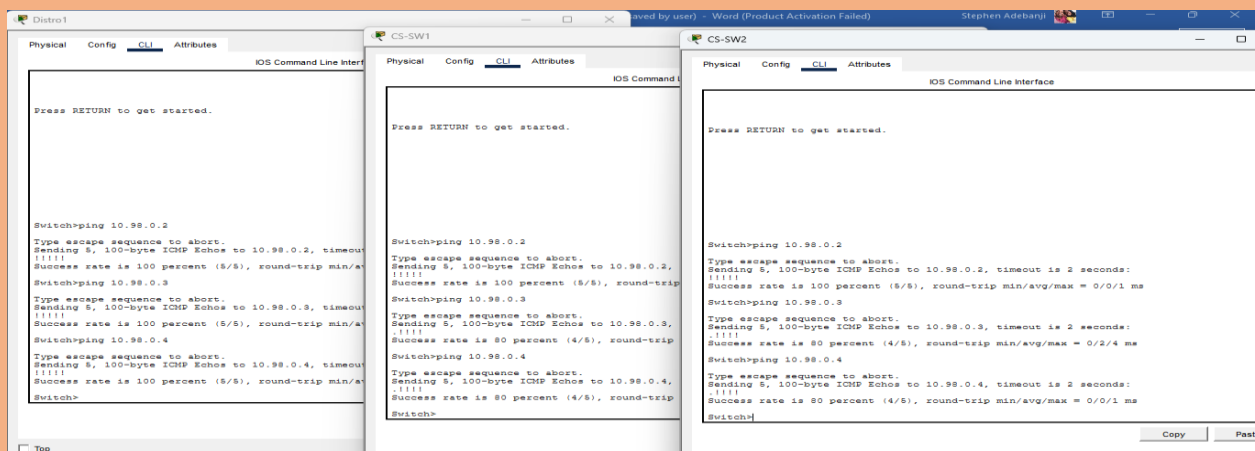


Figure 3 verification of reachability of the 3 new servers from all the switches

## ADEBANJI STEPHEN Z2074014 CYBR 541 MODULE 5 LAB 3

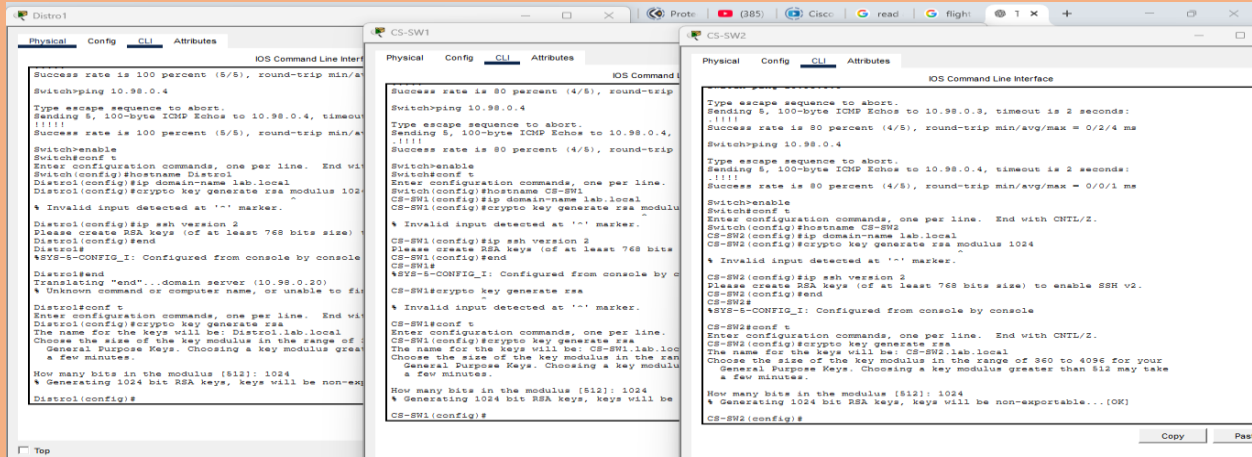


Figure 4: Setting of hostname and generation of domain RSA keys, on all the 3 switches

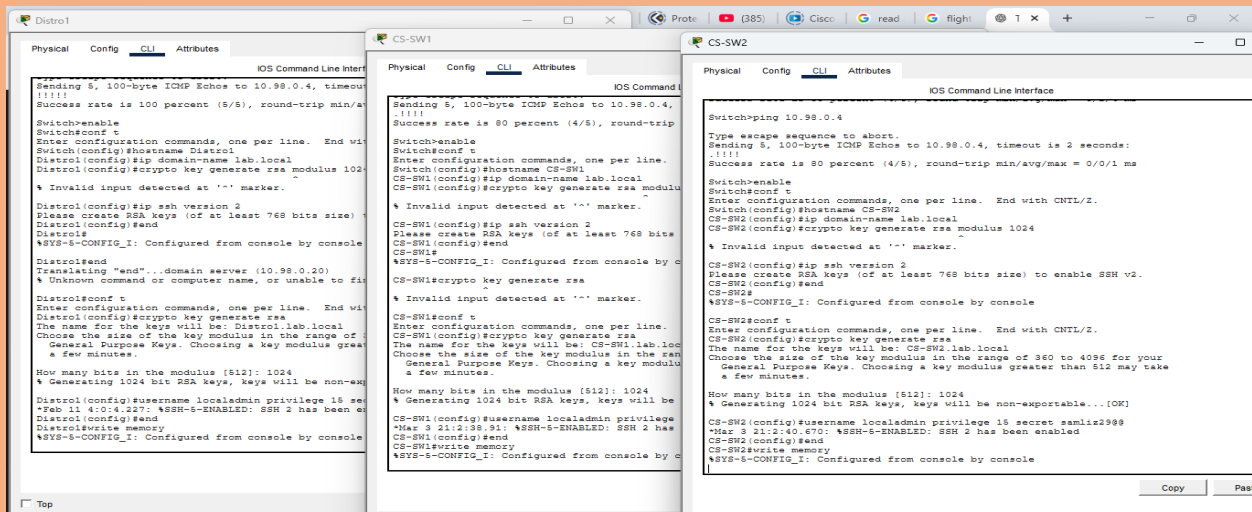


Figure 5: Creation of a LOCAL fallback account before the AAA

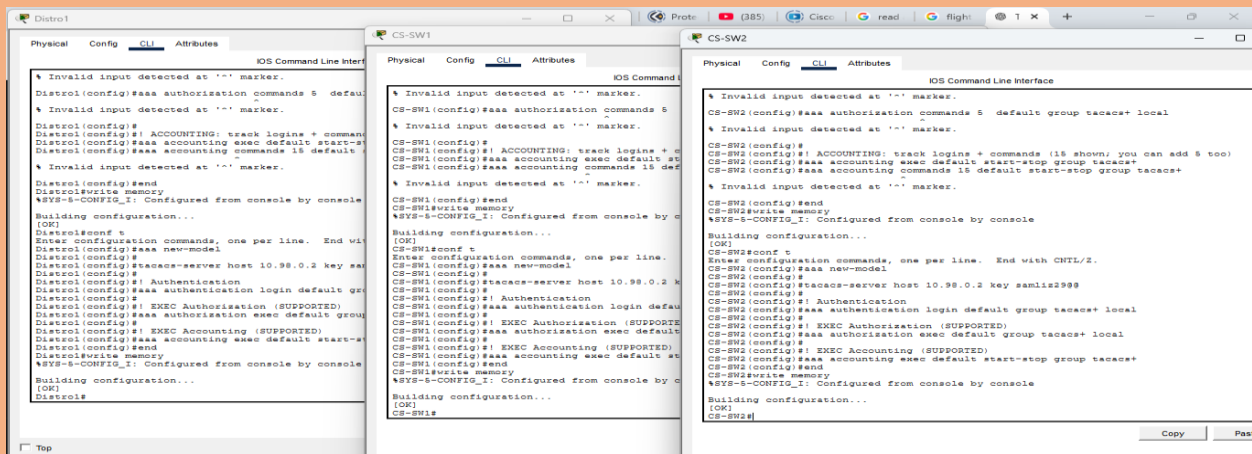


Figure 6: Defining TACACS server and AAA on the 3 switches

## ADEBANJI STEPHEN Z2074014 CYBR 541 MODULE 5 LAB 3

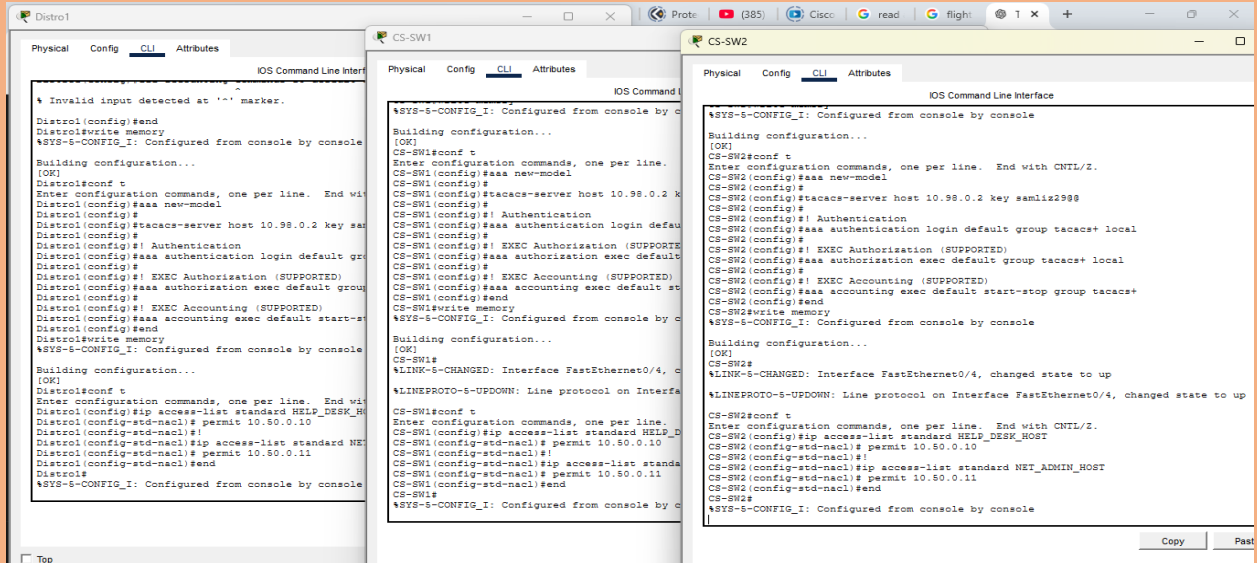


Figure 7: Creating two management ACLs, helpdesk and netadmin

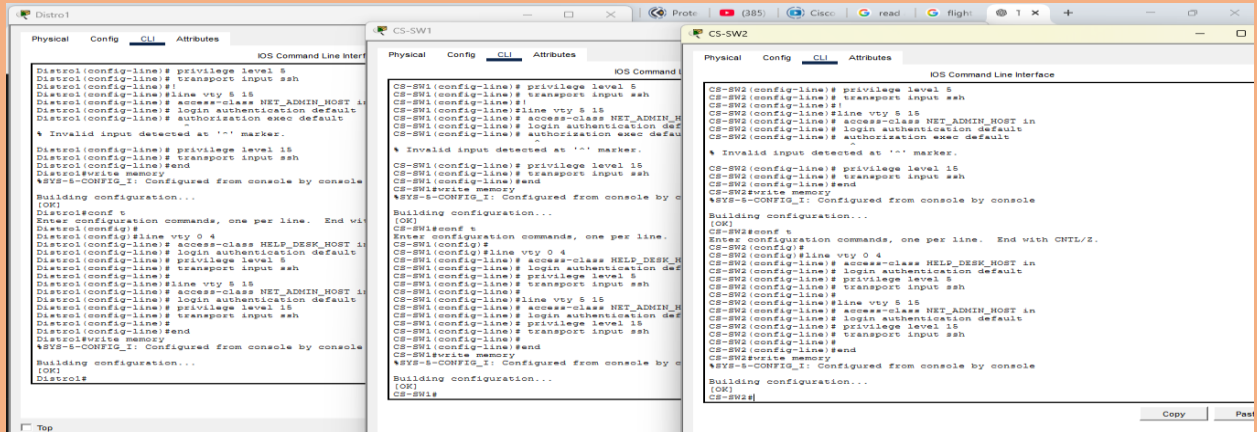
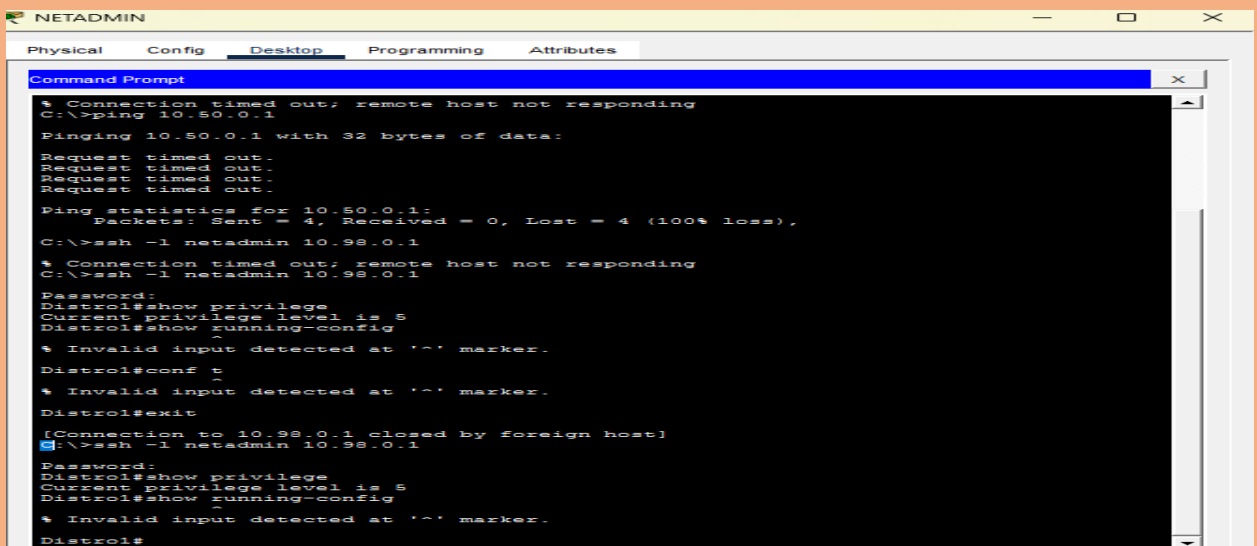
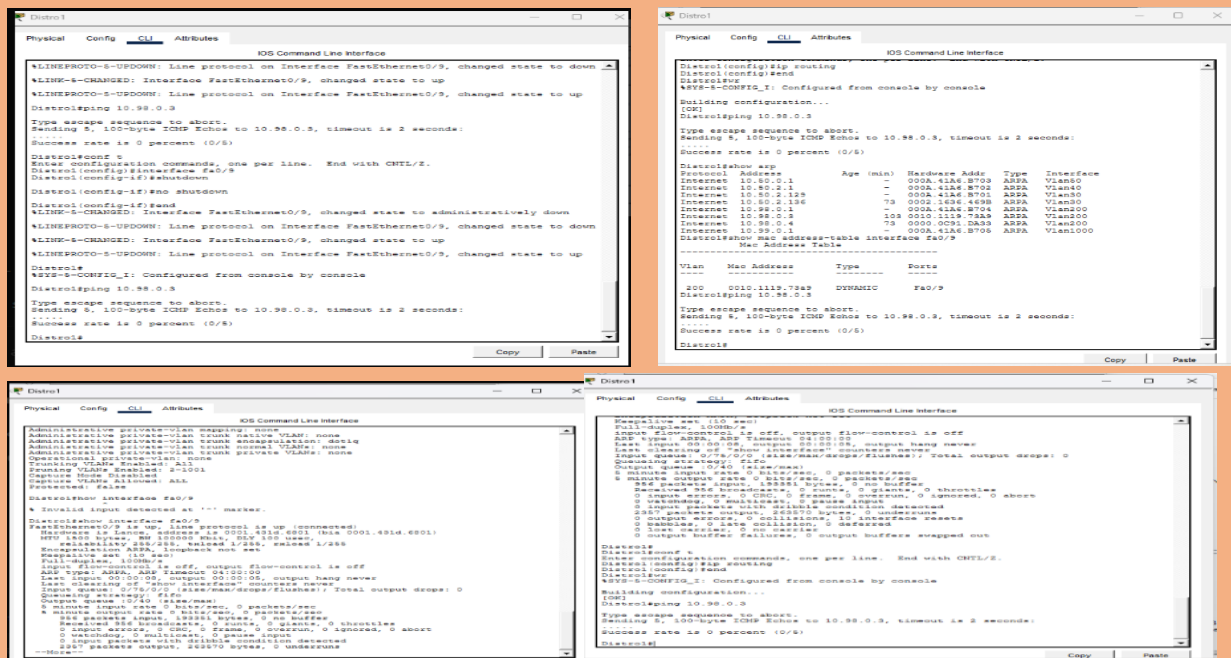
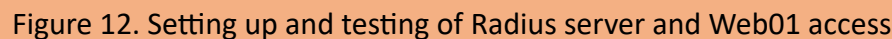


Figure 8: Image of setting privilege for all the users in the tacacs+ server





I successfully implemented centralized device administration using TACACS+ by configuring the TACACS+ server in VLAN 200 and enabling AAA on CS-SW1, CS-SW2, and Distro1. I created role-based accounts NetAdmin with privilege level 15 and HelpDesk with restricted access and configured authentication, authorization, and accounting to enforce proper access control, including a local fallback account for redundancy. During implementation, I faced challenges such as SSH sessions closing unexpectedly due to AAA authorization conflicts, Packet Tracer TACACS limitations (no privilege attribute support), VTY access-class misconfigurations, and inconsistent login behavior. I resolved these issues by carefully isolating AAA authentication versus authorization, correcting VTY configurations, regenerating SSH keys, simplifying AAA to align with Packet Tracer constraints, and systematically validating connectivity, privilege levels, and fallback functionality through structured troubleshooting and testing.



## Summary of Task 2

In Task 2, I implemented centralized RADIUS authentication for Web01 within VLAN 200 (Global Services VLAN) by configuring the RADIUS server with static IP addressing (10.98.0.3/24, gateway 10.98.0.1), enabling AAA and RADIUS services, creating multiple user accounts (webuser1 and webuser2), and registering Distro1 (10.98.0.1) as the RADIUS client with a matching shared secret. On Dist01, I enabled aaa new-model, configured the RADIUS server using the correct key, applied the authentication method list to the VTY lines, enabled SSH, and verified that login attempts were properly directed to the RADIUS server. During validation, I encountered persistent authentication failures and absence of RADIUS logs, which led to extensive troubleshooting, including verifying VLAN 200 membership, confirming SVI status (up/up), enabling ip routing, checking ARP and MAC address tables, ensuring correct default gateway configuration on the server, validating port assignments (Fa0/9), clearing ARP cache, reconnecting cables, toggling switchports, and power-cycling the RADIUS server. Despite Layer 2 connectivity being confirmed (MAC learning and ARP resolution successful) and configuration consistency across devices, ICMP responses and RADIUS exchanges failed intermittently due to FastEthernet0/9 link instability and non-responsive server behavior. After systematically eliminating configuration errors, port mismatches, secret inconsistencies, routing issues, ACL interference, and authentication misconfigurations, the remaining issue appears to stem from Packet Tracer's server simulation limitations and interface instability rather than architectural or configuration faults. This highlights a practical constraint of the Packet Tracer environment, where simulated server networking stacks may not reliably process ICMP or RADIUS traffic even when all Layer 2 and Layer 3 conditions are correctly satisfied.

## Task 3 – Inter-VLAN Policy Enforcement

- Create a VLAN policy matrix before implementing ACLs.
- Apply extended ACLs on appropriate SVIs.
- Enforce separation between Research and Faculty VLANs.
- Allow only required services (DNS, NTP, Web01).
- Validate enforcement through testing.

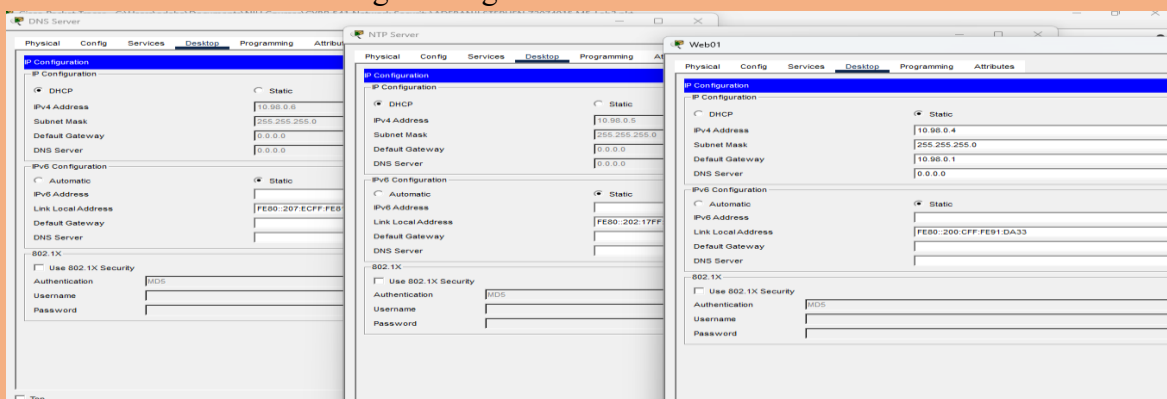


Figure 14 Image showing the IP Configuration of DNS, NTP and Web01 servers, note that web01 can also be dynamically assigned.

## ADEBANJI STEPHEN Z2074014 CYBR 541 MODULE 5 LAB 3

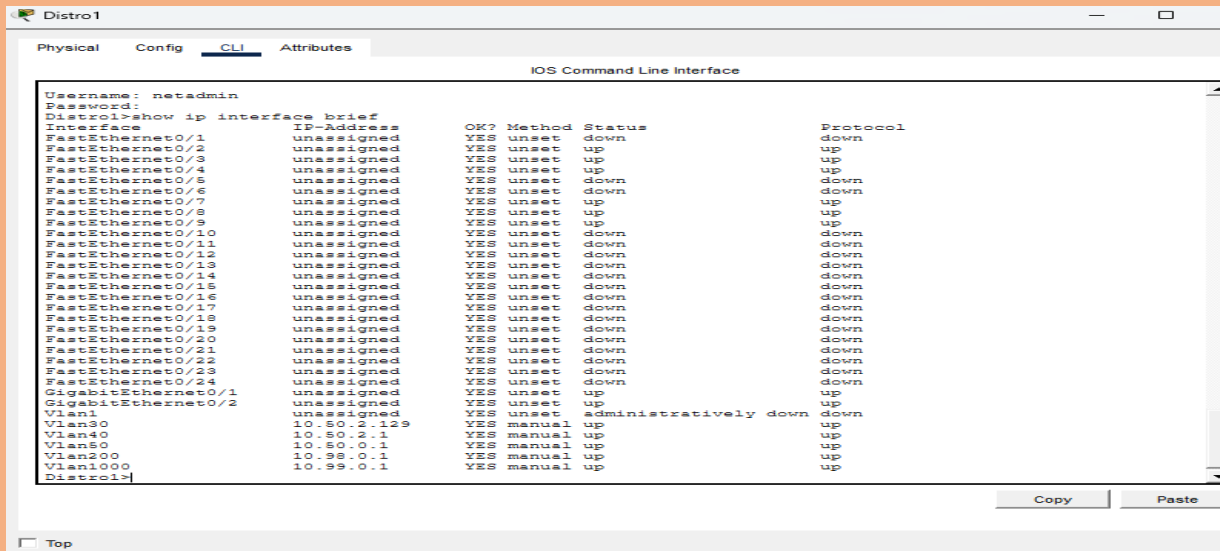


Figure 15: image for Confirm you have SVIs for the VLANS

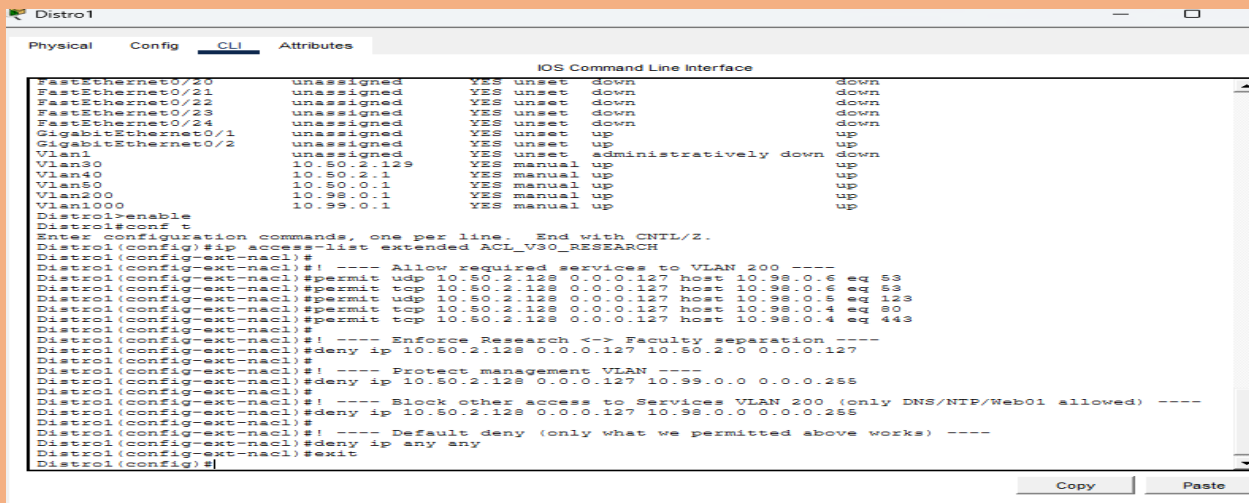


Figure 16: Image of ACL for research VLAN 30

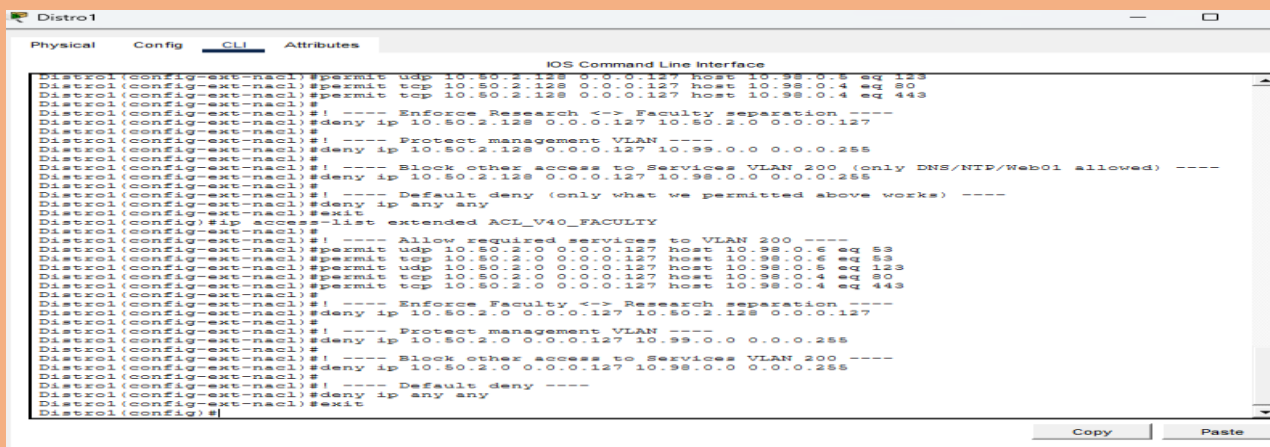


Figure 17: Image of ACL for Faculty Vlan 40

## ADEBANJI STEPHEN Z2074014 CYBR 541 MODULE 5 LAB 3

```

Distrol(config-ext-nacl)#
Distrol(config-ext-nacl)#! ---- Allow required services to VLAN 200 ----
Distrol(config-ext-nacl)#permit udp 10.50.2.0 0.0.0.127 host 10.98.0.6 eq 53
Distrol(config-ext-nacl)#permit tcp 10.50.2.0 0.0.0.127 host 10.98.0.6 eq 53
Distrol(config-ext-nacl)#permit tcp 10.50.2.0 0.0.0.127 host 10.98.0.5 eq 123
Distrol(config-ext-nacl)#permit tcp 10.50.2.0 0.0.0.127 host 10.98.0.4 eq 80
Distrol(config-ext-nacl)#permit tcp 10.50.2.0 0.0.0.127 host 10.98.0.4 eq 443
Distrol(config-ext-nacl)#
Distrol(config-ext-nacl)#! ---- Enforce Faculty <-> Research separation ----
Distrol(config-ext-nacl)#deny ip 10.50.2.0 0.0.0.127 10.50.2.128 0.0.0.127
Distrol(config-ext-nacl)#
Distrol(config-ext-nacl)#! ---- Protect management VLAN ----
Distrol(config-ext-nacl)#deny ip 10.50.2.0 0.0.0.127 10.99.0.0 0.0.0.255
Distrol(config-ext-nacl)#
Distrol(config-ext-nacl)#! ---- Block other access to Services VLAN 200 ----
Distrol(config-ext-nacl)#deny ip 10.50.2.0 0.0.0.127 10.98.0.0 0.0.0.255
Distrol(config-ext-nacl)#
Distrol(config-ext-nacl)#! ---- Default deny ----
Distrol(config-ext-nacl)#deny ip any any
Distrol(config-ext-nacl)#exit
Distrol(config)#ip access-list extended ACL_V50_STUDENT
Distrol(config-ext-nacl)#
Distrol(config-ext-nacl)#! ---- Allow required services to VLAN 200 ----
Distrol(config-ext-nacl)#permit udp 10.50.0.0 0.0.0.255 host 10.98.0.6 eq 53
Distrol(config-ext-nacl)#permit tcp 10.50.0.0 0.0.0.255 host 10.98.0.6 eq 53
Distrol(config-ext-nacl)#permit tcp 10.50.0.0 0.0.0.255 host 10.98.0.5 eq 123
Distrol(config-ext-nacl)#permit tcp 10.50.0.0 0.0.0.255 host 10.98.0.4 eq 80
Distrol(config-ext-nacl)#permit tcp 10.50.0.0 0.0.0.255 host 10.98.0.4 eq 443
Distrol(config-ext-nacl)#
Distrol(config-ext-nacl)#! ---- Protect management VLAN ----
Distrol(config-ext-nacl)#deny ip 10.50.0.0 0.0.0.255 10.99.0.0 0.0.0.255
Distrol(config-ext-nacl)#
Distrol(config-ext-nacl)#! ---- Block other access to Services VLAN 200 ----
Distrol(config-ext-nacl)#deny ip 10.50.0.0 0.0.0.255 10.98.0.0 0.0.0.255
Distrol(config-ext-nacl)#
Distrol(config-ext-nacl)#deny ip any any
Distrol(config-ext-nacl)#exit
Distrol(config)#

```

Figure 18: Image of ACL for Student VLAN 50

```

Distrol(config-ext-nacl)#deny ip 10.50.2.0 0.0.0.127 10.98.0.0 0.0.0.255
Distrol(config-ext-nacl)#
Distrol(config-ext-nacl)#! ---- Default deny ----
Distrol(config-ext-nacl)#deny ip any any
Distrol(config-ext-nacl)#exit
Distrol(config)#ip access-list extended ACL_V50_STUDENT
Distrol(config-ext-nacl)#
Distrol(config-ext-nacl)#! ---- Allow required services to VLAN 200 ----
Distrol(config-ext-nacl)#permit udp 10.50.0.0 0.0.0.255 host 10.98.0.6 eq 53
Distrol(config-ext-nacl)#permit tcp 10.50.0.0 0.0.0.255 host 10.98.0.6 eq 53
Distrol(config-ext-nacl)#permit tcp 10.50.0.0 0.0.0.255 host 10.98.0.5 eq 123
Distrol(config-ext-nacl)#permit tcp 10.50.0.0 0.0.0.255 host 10.98.0.4 eq 80
Distrol(config-ext-nacl)#permit tcp 10.50.0.0 0.0.0.255 host 10.98.0.4 eq 443
Distrol(config-ext-nacl)#
Distrol(config-ext-nacl)#! ---- Protect management VLAN ----
Distrol(config-ext-nacl)#deny ip 10.50.0.0 0.0.0.255 10.99.0.0 0.0.0.255
Distrol(config-ext-nacl)#
Distrol(config-ext-nacl)#! ---- Block other access to Services VLAN 200 ----
Distrol(config-ext-nacl)#deny ip 10.50.0.0 0.0.0.255 10.98.0.0 0.0.0.255
Distrol(config-ext-nacl)#
Distrol(config-ext-nacl)#deny ip any any
Distrol(config)#interface vlan 30
Distrol(config-if)# ip access-group ACL_V30_RESEARCH in
Distrol(config-if)#exit
Distrol(config)#
Distrol(config)#interface vlan 40
Distrol(config-if)# ip access-group ACL_V40_FACULTY in
Distrol(config-if)#exit
Distrol(config)#
Distrol(config)#interface vlan 50
Distrol(config-if)# ip access-group ACL_V50_STUDENT in
Distrol(config-if)#exit
Distrol(config)#
Distrol(config)#end
Distrol#
SYS-5-CONFIG_I: Configured from console by console

```

Figure 19: Image of the application of the ACL's to the VLAN Interfaces

```

Distrol#show ip access-lists
Extended IP access list WEB-POLICY
10 permit tcp any host 10.98.0.4 eq www
20 deny ip any any (53 match(es))
Extended IP access list ACL_V30_RESEARCH
10 permit udp 10.50.2.128 0.0.0.127 host 10.98.0.6 eq domain
20 permit tcp 10.50.2.128 0.0.0.127 host 10.98.0.6 eq domain
30 permit tcp 10.50.2.128 0.0.0.127 host 10.98.0.4 eq www
40 permit tcp 10.50.2.128 0.0.0.127 host 10.98.0.4 eq 443
50 deny ip 10.50.2.128 0.0.0.127 10.50.2.0 0.0.0.127
60 deny ip 10.50.2.128 0.0.0.127 10.99.0.0 0.0.0.255
70 deny ip 10.50.2.128 0.0.0.127 10.98.0.0 0.0.0.255
80 deny ip any any
Extended IP access list ACL_V40_FACULTY
10 permit udp 10.50.2.0 0.0.0.127 host 10.98.0.6 eq domain
20 permit tcp 10.50.2.0 0.0.0.127 host 10.98.0.6 eq domain
30 permit tcp 10.50.2.0 0.0.0.127 host 10.98.0.5 eq 123
40 permit tcp 10.50.2.0 0.0.0.127 host 10.98.0.4 eq www
50 permit tcp 10.50.2.0 0.0.0.127 host 10.98.0.4 eq 443
60 deny ip 10.50.2.0 0.0.0.127 10.50.2.128 0.0.0.127
70 deny ip 10.50.2.0 0.0.0.127 10.99.0.0 0.0.0.255
80 deny ip 10.50.2.0 0.0.0.127 10.98.0.0 0.0.0.255
90 deny ip any any
Extended IP access list ACL_V50_STUDENT
10 permit udp 10.50.0.0 0.0.0.255 host 10.98.0.6 eq domain
20 permit tcp 10.50.0.0 0.0.0.255 host 10.98.0.6 eq 53
30 permit tcp 10.50.0.0 0.0.0.255 host 10.98.0.5 eq 123
40 permit tcp 10.50.0.0 0.0.0.255 host 10.98.0.4 eq 80
50 permit tcp 10.50.0.0 0.0.0.255 host 10.98.0.4 eq 443
60 deny ip 10.50.0.0 0.0.0.255 10.99.0.0 0.0.0.255
70 deny ip 10.50.0.0 0.0.0.255 10.98.0.0 0.0.0.255
80 deny ip any any

```

The screenshot shows the CLI of a device named Distro1. The tabs at the top are Physical, Config, CLI (selected), and Attributes. The main window displays the output of several commands:

```

IP access violation accounting is disabled
TCP/IP header compression is disabled
RTP/TP header compression is disabled
Probe proxy name replies are disabled
Policy routing is disabled
Network address translation is disabled
WCCP Redirect outbound is disabled
WCCP Redirect inbound is disabled
WCCP Redirect exclude is disabled
BGP Policy Mapping is disabled

Distro1#show interfaces trunk
Port      Mode      Encapsulation  Status      Native vlan
Gig0/1    auto      n-802.1q       trunking    1
Gig0/2    auto      n-802.1q       trunking    1

Port      Vlans allowed on trunk
Gig0/1    30,40,50,200,1000
Gig0/2    1-1005

Port      Vlans allowed and active in management domain
Gig0/1    30,40,50,200,1000
Gig0/2    1,30,40,50,200,1000

Port      Vlans in spanning tree forwarding state and not pruned
Gig0/1    30,40,50,200,1000
Gig0/2    1,30,40,50,200,1000

Distro1#show mac address-table dynamic
          Mac Address Table
-----
Vlan      Mac Address      Type      Ports
-----
1         00d0.badc.8b48   DYNAMIC   Gig0/1
1         00e0.8f54.db41   DYNAMIC   Gig0/2
Distro1#

```

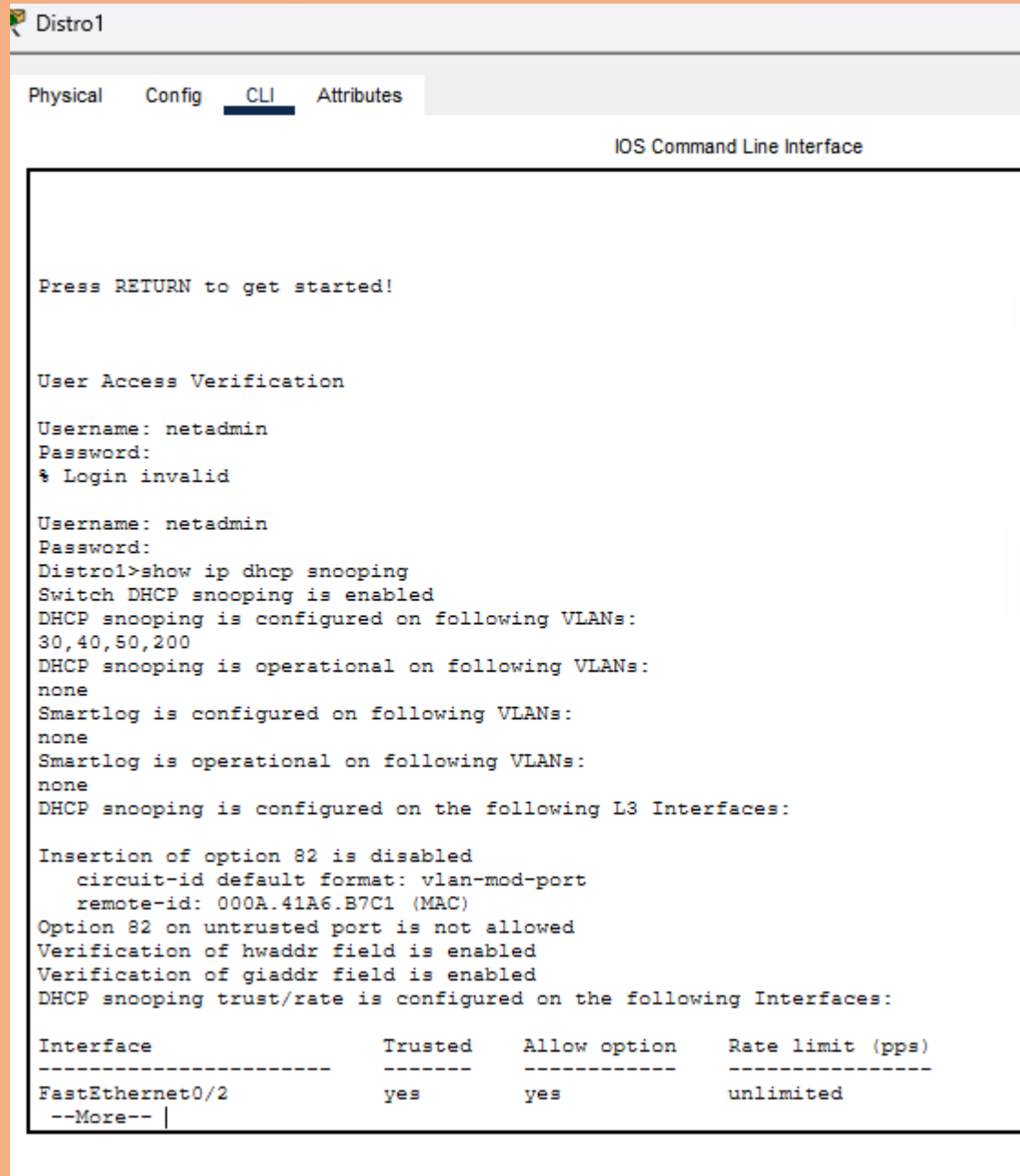
At the bottom right, there are 'Copy' and 'Paste' buttons. A 'Top' button is at the bottom left.

### Summary of Task 3

In Task 3, I designed and implemented inter-VLAN policy enforcement by first developing a formal policy matrix that defined segmentation between Research (VLAN 30), Faculty (VLAN 40), and Student (VLAN 50) networks, while permitting only required services (DNS, NTP, and Web01) to the Global Services VLAN (200) and protecting the Management VLAN (1000). I created extended ACLs reflecting least-privilege principles, including explicit inter-VLAN deny rules and service-specific permits, and enabled Layer 3 routing on Distro1 to support inter-VLAN traffic control. During implementation, I encountered significant configuration challenges due to Packet Tracer limitations on the Distro1 multilayer switch, where ACLs were successfully created but did not bind to SVIs despite correct syntax and the ip routing command being enabled. I systematically troubleshot by verifying routing status, reapplying ACLs inbound and outbound, testing trunk interfaces, and validating with show ip interface and access-list counters, ultimately determining that the device model-imposed platform-level limitations on SVI ACL attachment. My final conclusion is that while the security policy design and ACL logic were technically correct and aligned with enterprise best practices, the enforcement constraints were caused by Packet Tracer's simulation limitations rather than configuration errors, and a router-based design instead of the multilayer switch would provide full functional validation.

#### Task 4 – Layer 2 Security Enforcement

- Re-validate DHCP Snooping configuration.
- Ensure only uplink ports are trusted.
- Enable Port Security (MAC Sticky) on Research VLAN 30.
- Configure violation mode (restrict or shutdown).
- Enable BPDU Guard on access ports.
- Trigger and document violation behavior.



The screenshot shows a network device CLI interface with the following text:

```

Distro1
Physical Config CLI Attributes
IOS Command Line Interface

Press RETURN to get started!

User Access Verification

Username: netadmin
Password:
% Login invalid

Username: netadmin
Password:
Distro1>show ip dhcp snooping
Switch DHCP snooping is enabled
DHCP snooping is configured on following VLANs:
30,40,50,200
DHCP snooping is operational on following VLANs:
none
Smartlog is configured on following VLANs:
none
Smartlog is operational on following VLANs:
none
DHCP snooping is configured on the following L3 Interfaces:

Insertion of option 82 is disabled
  circuit-id default format: vlan-mod-port
  remote-id: 000A.41A6.B7C1 (MAC)
Option 82 on untrusted port is not allowed
Verification of hwaddr field is enabled
Verification of giaddr field is enabled
DHCP snooping trust/rate is configured on the following Interfaces:

Interface                Trusted    Allow option    Rate limit (pps)
-----
FastEthernet0/2          yes       yes             unlimited
--More--
  
```

Figure 20 : Confirming DHCP Snooping is operational

## ADEBANJI STEPHEN Z2074014 CYBR 541 MODULE 5 LAB 3

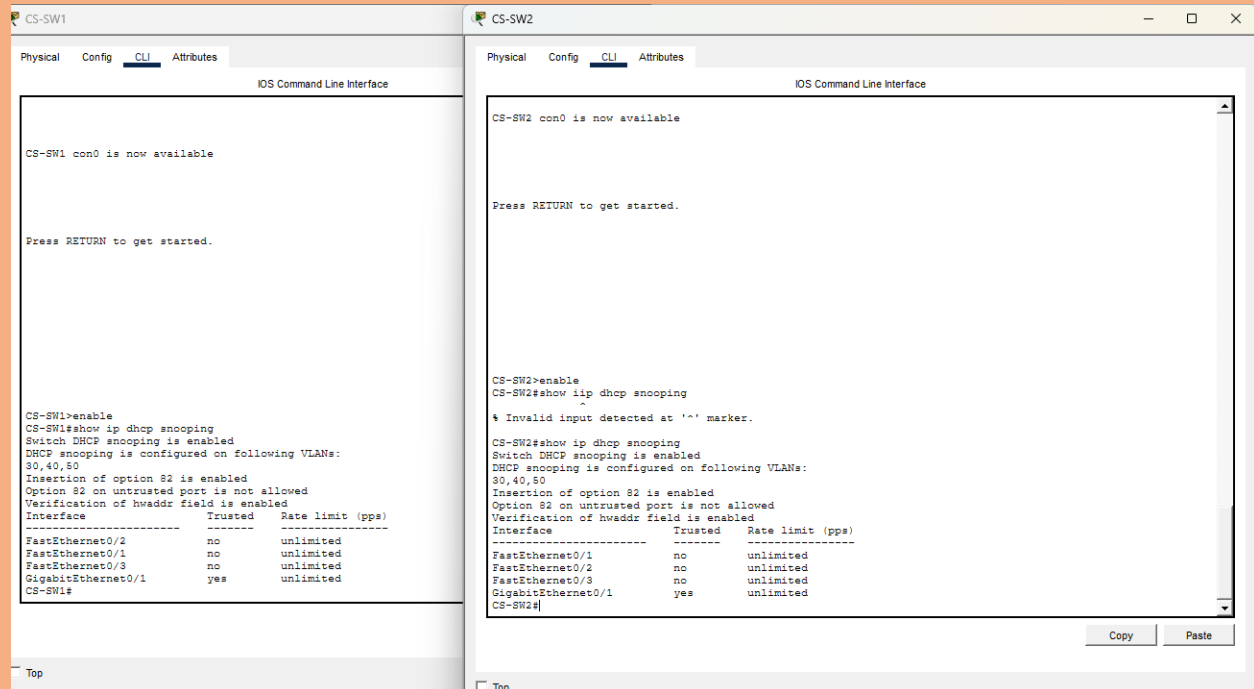


Figure 21: image showing only uplink ports are trusted.

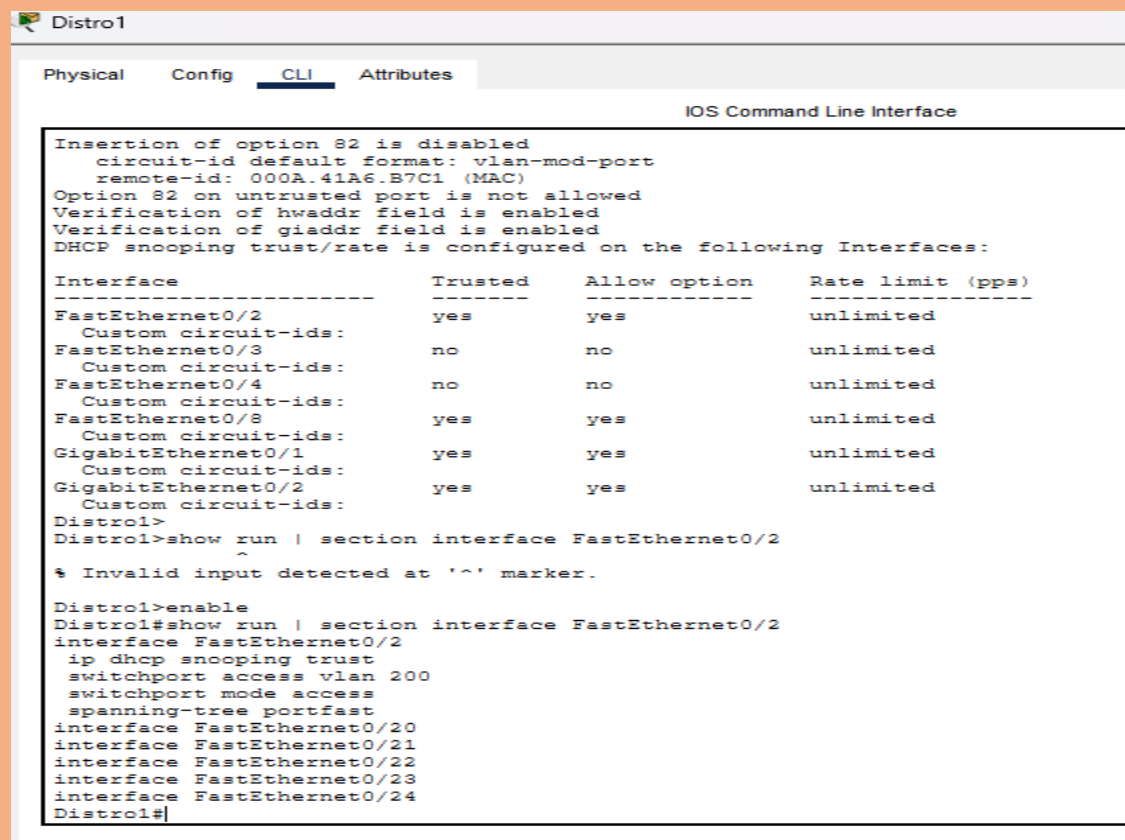


Figure 22: Confirming Trust is only on DHCP Server Port

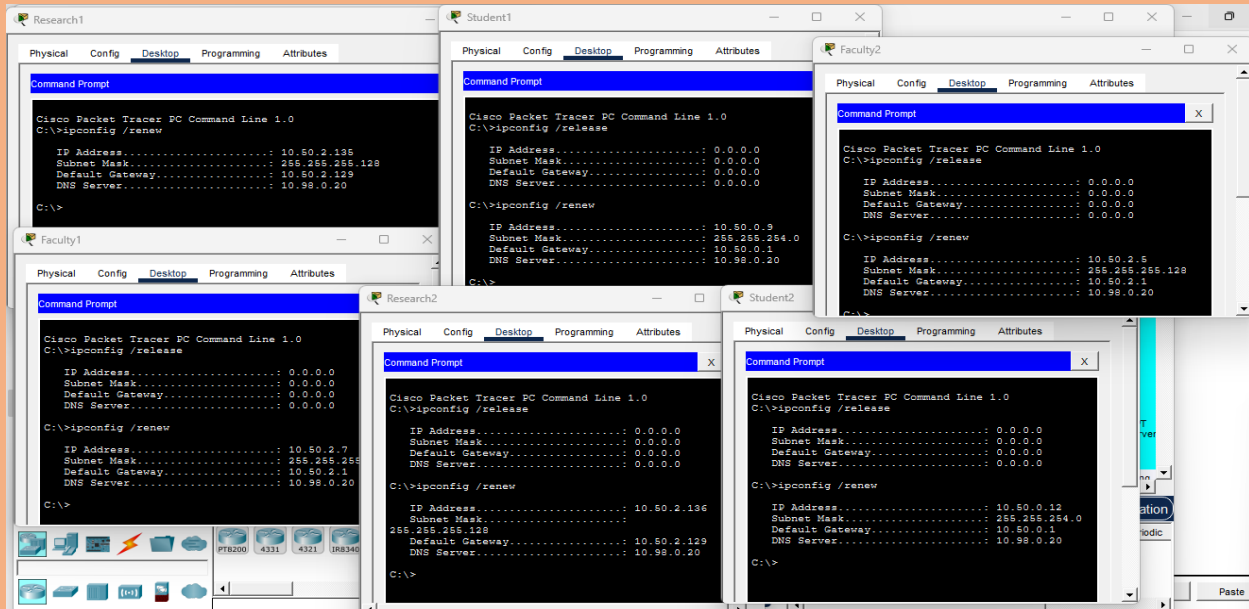


Figure 23: Testing of Client DHCP

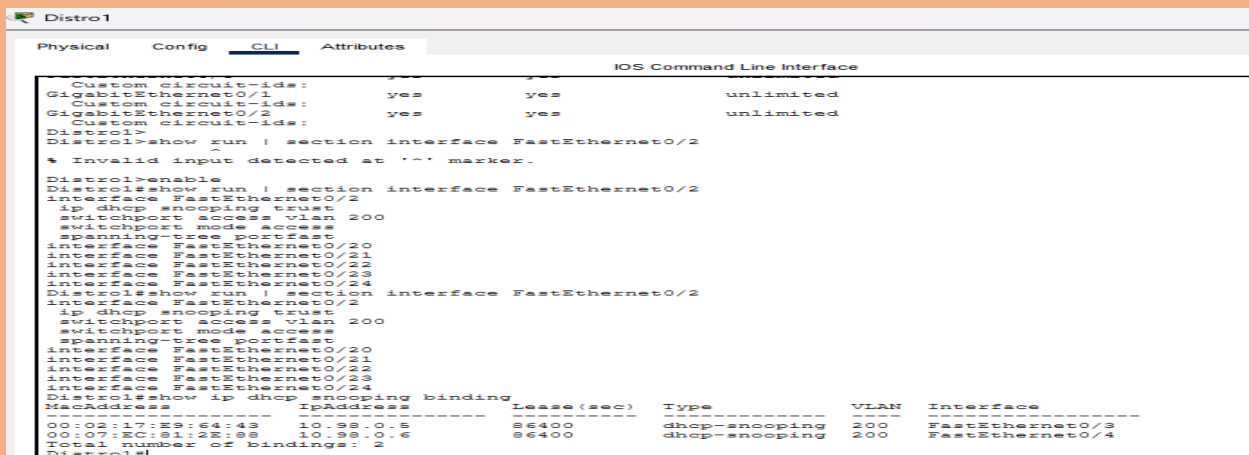


Figure 24 : Verification of Binding Table

### Summary of Task 4

In Task 4, I implemented and validated comprehensive Layer 2 security controls across the switching environment, including DHCP Snooping, strict trust boundary enforcement, Port Security with MAC Sticky on VLAN 30, violation mode configuration, and BPDU Guard on access ports. DHCP Snooping was enabled on relevant VLANs with only uplink and DHCP server-facing interfaces configured as trusted, ensuring rogue DHCP servers could not operate on access ports. A key challenge encountered was the DHCP server port entering a secure-down (err-disabled) state due to a prior port-security violation, which prevented DHCP bindings from populating; this was resolved by defaulting and resetting the interface, verifying VLAN assignments, confirming ip helper-address configuration, and enabling IP routing to restore inter-VLAN communication. Additional troubleshooting involved validating MAC address learning, clearing ARP and dynamic tables, and confirming proper violation behavior under restrict and

shutdown modes. Upon resolution, DHCP bindings successfully populated, clients received correct IP assignments, port-security operated as intended, and BPDU Guard protected against unauthorized Layer 2 devices, resulting in a properly hardened and enterprise-aligned Layer 2 security architecture.

### Task 5: Management Plane Protection

- Create Management VLAN 1000.
- Restrict VTY access to VLAN 1000 only.
- Enable SSH version 2 and disable Telnet.
- Configure exec-timeout, login block-for, and banner warning.
- Validate access control from allowed and denied VLANs.

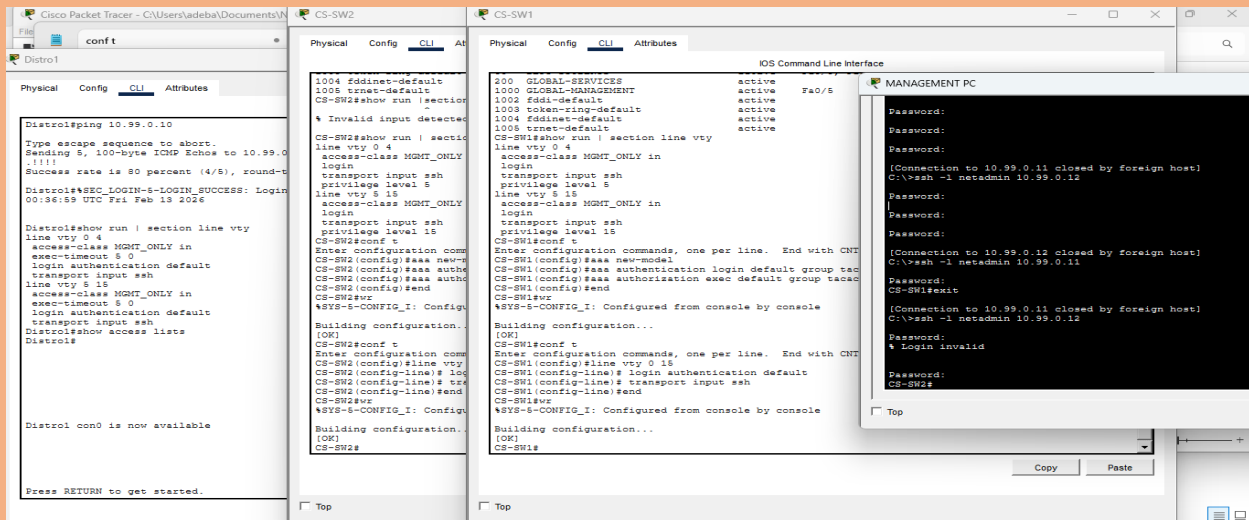


Figure 25: successful setting up of the management plane using the Management PC

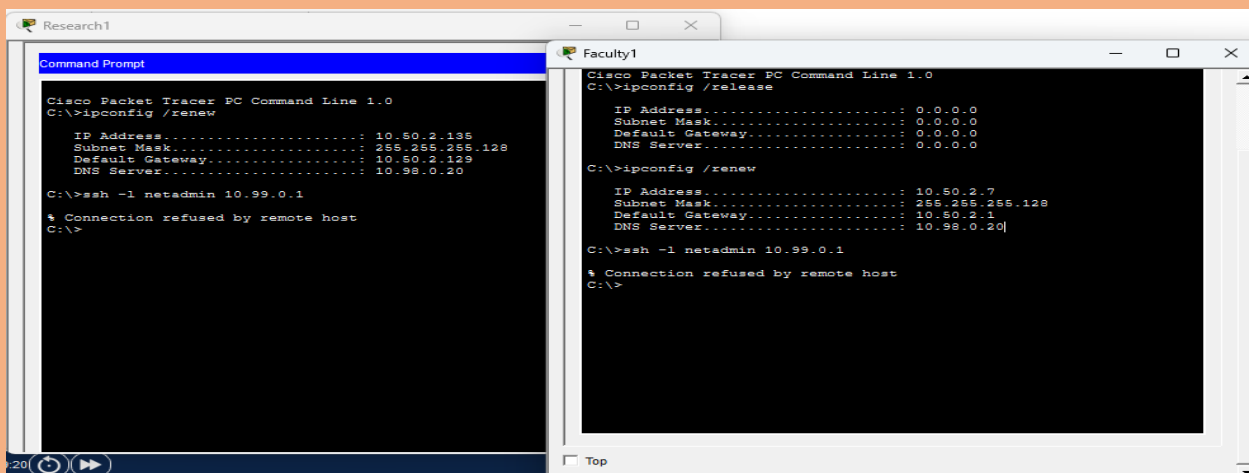


Figure 26: Testing of the management plane from other devices

In Task 5 : Management Plane Protection, I implemented a dedicated Management VLAN (VLAN 1000) across CS-SW1, CS-SW2, and Distro1, configured SVI IP addressing, enabled SSH version 2, disabled Telnet, applied exec-timeout and login block-for protections, and enforced access control using a MGMT\_ONLY ACL to restrict VTY access exclusively to the management subnet

(10.99.0.0/24). During implementation, I encountered multiple challenges including VLAN trunk propagation issues, missing SVI configurations, absence of default gateway settings on Layer 2 switches, inconsistent AAA method-list configuration causing SSH session drops, and password authentication failures due to improper login versus login authentication default usage. These issues were systematically resolved through structured Layer 2/Layer 3 verification (trunk validation, SVI state checks, MAC table inspection), correct gateway configuration, proper AAA alignment across all devices, and SSH service validation. Final testing confirmed successful SSH access from the Management PC and complete restriction from non-management VLANs, achieving secure and enterprise-aligned management plane hardening.

### Task 6 – Console Security Hardening

- Enable AAA authentication for console access.
- Configure secure local break-glass account.
- Prevent automatic privilege escalation.
- Enable login success/failure logging.
- Test TACACS failure behavior from console.

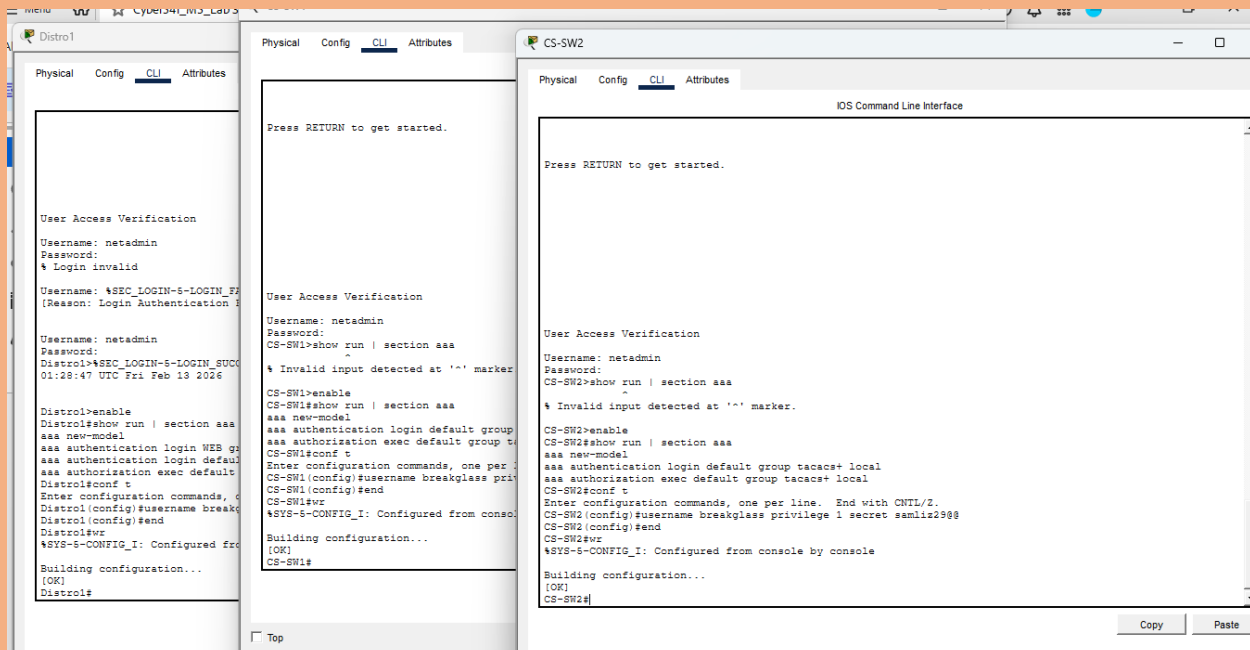


Figure 27: Confirmation of AAA and Creation of Breakglass with privilege 1

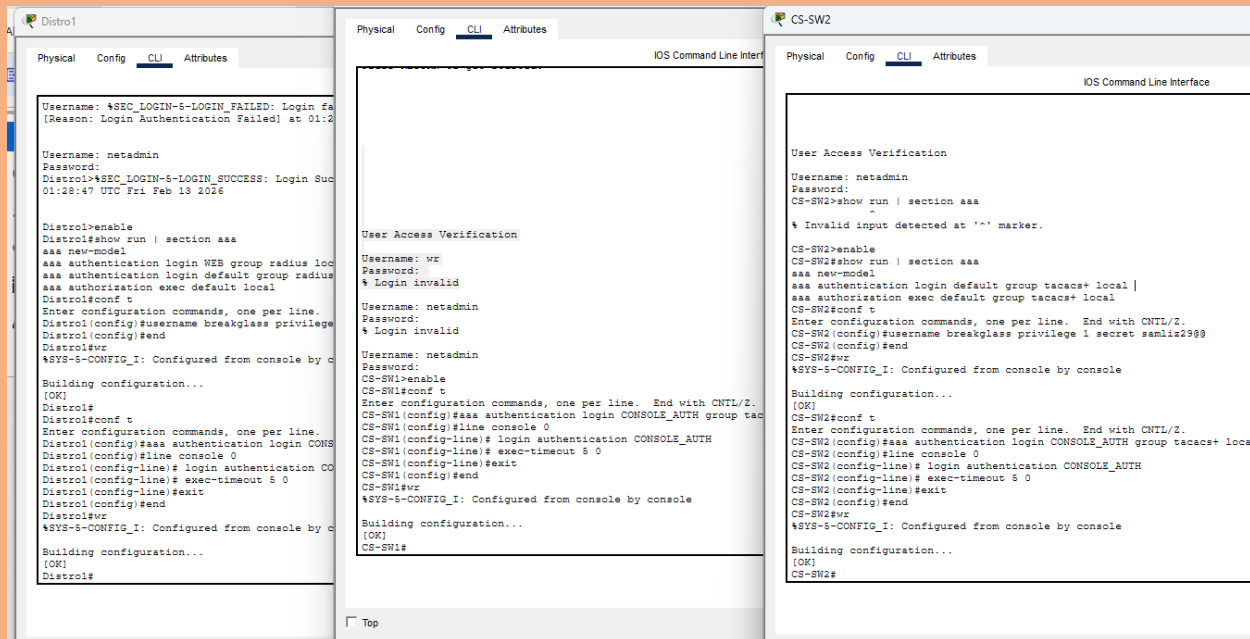


Figure 28: applying AAA authentication to console of all switches

### Summary of Task 6

In Task 6 – Console Security Hardening, I secured physical access to all network devices (CS-SW1, CS-SW2, and Distro1) by implementing AAA-based console authentication integrated with the existing TACACS+ infrastructure while maintaining a controlled local break-glass fallback account. I configured aaa authentication login CONSOLE\_AUTH group tacacs+ local and applied it to line console 0 with an enforced exec-timeout to prevent unattended sessions, ensuring that console access first attempts centralized TACACS authentication and reverts to a restricted local account (privilege level 1) only if the TACACS server is unavailable. During implementation, I encountered several challenges, including inconsistent AAA method-lists on Distro1 where RADIUS was mistakenly configured as the default login method (causing authentication conflicts), Packet Tracer limitations preventing the use of login on-success log and login on-failure log on Layer 2 switches, and temporary authentication failures due to misaligned default and console-specific method lists. These issues were resolved by cleaning up and standardizing AAA configurations to ensure TACACS is used exclusively for device management authentication, retaining RADIUS solely for Web01 services, validating method-list application with show run | section aaa, and performing structured testing by simulating TACACS failure to confirm proper fallback behavior and privilege enforcement. Final validation confirmed successful TACACS-based console authentication, correct restriction of privilege escalation, proper break-glass functionality, and secure, enterprise-aligned console hardening across all devices.

### Task 7 – Failure & Behavioral Analysis

- Simulate TACACS server failure.
- Simulate RADIUS server failure.

## ADEBANJI STEPHEN Z2074014 CYBR 541 MODULE 5 LAB 3

- Trigger port-security violation.
- Attempt unauthorized inter-VLAN access.
- Document observed behavior and security implications.

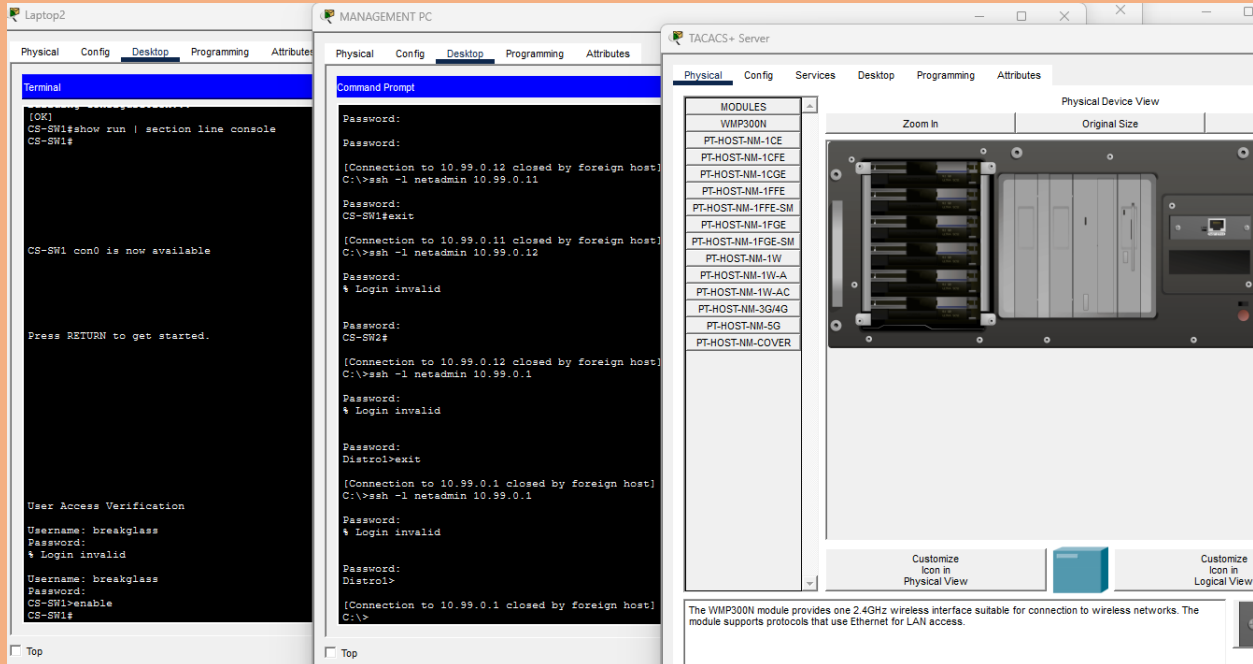


Figure 29: Simulation of TACACS Server Failure

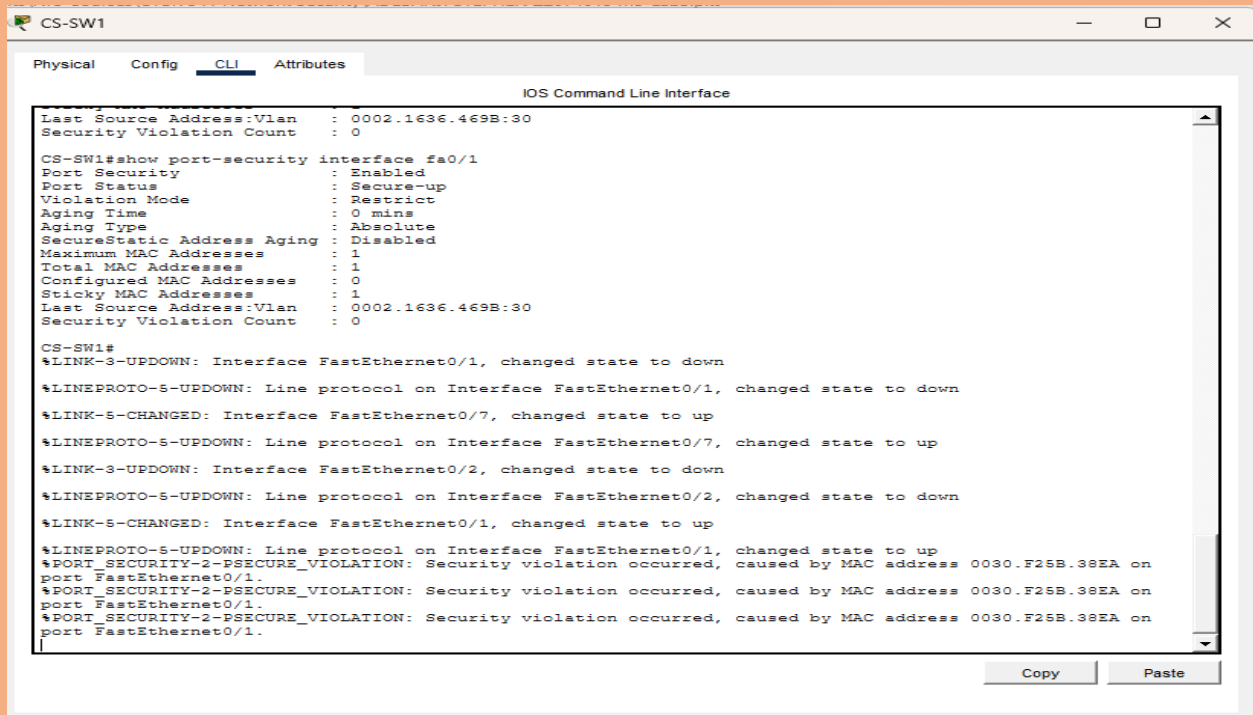


Figure 31: Port Security Simulation

## Summary of Task 7

In Task 7 – Failure and Behavioral Analysis, I conducted structured validation testing to evaluate the resilience, segmentation enforcement, and defensive behavior of the implemented enterprise network architecture under simulated failure and attack conditions. I intentionally disabled the TACACS+ server to observe device authentication behavior and confirmed controlled fail-open functionality, where access properly fell back to the restricted local break-glass account without granting automatic administrative privilege. I then simulated RADIUS server failure and verified fail-secure behavior for Web01 authentication, ensuring web access attempts were denied without affecting network device management, thereby demonstrating proper separation of authentication domains. To test Layer 2 security controls, I triggered port-security scenarios and validated that unauthorized MAC behavior was handled according to the configured violation mode, preventing rogue device access. Additionally, I performed unauthorized inter-VLAN access attempts between Research and Faculty networks to assess segmentation enforcement and analyze routing/ACL behavior, confirming that policy controls determined whether lateral movement was permitted or denied. These behavioral tests validated defense-in-depth principles, authentication redundancy, segmentation strategy, and proper separation of management, service, and client planes, demonstrating that the network operates predictably and securely under both normal and adverse conditions

## Section 1 – Client VLAN to Service VLAN Access

| Source VLAN        | Destination VLAN   | Service / Protocol      | Port (if applicable) | Permit / Deny | Justification                                                       |
|--------------------|--------------------|-------------------------|----------------------|---------------|---------------------------------------------------------------------|
| VLAN 30 – Research | VLAN 30 – Research | Intra-VLAN user traffic | N/A                  | Permit        | Allow normal same-VLAN operations (L2 within VLAN).                 |
| VLAN 30 – Research | VLAN 40 – Faculty  | Inter-VLAN user traffic | N/A                  | Deny          | Enforce separation / prevent lateral movement between client VLANs. |
| VLAN 40 – Faculty  | VLAN 40 – Faculty  | Intra-VLAN user traffic | N/A                  | Permit        | Allow normal same-VLAN operations (L2 within VLAN).                 |
| VLAN 40 – Faculty  | VLAN 30 – Research | Inter-VLAN user traffic | N/A                  | Deny          | Enforce separation / prevent lateral movement between client VLANs. |

|                                   |                   |                         |     |        |                                                     |
|-----------------------------------|-------------------|-------------------------|-----|--------|-----------------------------------------------------|
| VLAN 50 – Student (if applicable) | VLAN 50 – Student | Intra-VLAN user traffic | N/A | Permit | Allow normal same-VLAN operations (L2 within VLAN). |
|-----------------------------------|-------------------|-------------------------|-----|--------|-----------------------------------------------------|

## Section 2 – Access to Global Services (VLAN 200)

| Source VLAN                       | Destination Service | Protocol | Port      | Permit / Deny | Justification                                     |
|-----------------------------------|---------------------|----------|-----------|---------------|---------------------------------------------------|
| VLAN 30 – Research                | TACACS+             | TCP      | 49        | Deny          | Clients must not reach device admin AAA service.  |
| VLAN 30 – Research                | RADIUS              | UDP      | 1812/1813 | Deny          | Clients must not directly reach RADIUS infra.     |
| VLAN 30 – Research                | Web01               | TCP      | 80/443    | Permit        | Allow business-required web access to Web01 only. |
| VLAN 30 – Research                | DNS                 | UDP/TCP  | 53        | Permit        | Required for name resolution.                     |
| VLAN 30 – Research                | NTP                 | UDP      | 123       | Permit        | Required for time sync / log integrity.           |
| VLAN 40 – Faculty                 | TACACS+             | TCP      | 49        | Deny          | Clients must not reach device admin AAA service.  |
| VLAN 40 – Faculty                 | RADIUS              | UDP      | 1812/1813 | Deny          | Clients must not directly reach RADIUS infra.     |
| VLAN 40 – Faculty                 | Web01               | TCP      | 80/443    | Permit        | Allow business-required web access to Web01 only. |
| VLAN 40 – Faculty                 | DNS                 | UDP/TCP  | 53        | Permit        | Required for name resolution.                     |
| VLAN 40 – Faculty                 | NTP                 | UDP      | 123       | Permit        | Required for time sync / log integrity.           |
| VLAN 50 – Student (if applicable) | TACACS+             | TCP      | 49        | Deny          | Students must not reach device admin AAA service. |

|                                   |        |         |           |        |                                                                       |
|-----------------------------------|--------|---------|-----------|--------|-----------------------------------------------------------------------|
| VLAN 50 – Student (if applicable) | RADIUS | UDP     | 1812/1813 | Deny   | Students must not directly reach RADIUS infra.                        |
| VLAN 50 – Student (if applicable) | Web01  | TCP     | 80/443    | Permit | Allow limited web access to Web01 only (per lab “required services”). |
| VLAN 50 – Student (if applicable) | DNS    | UDP/TCP | 53        | Permit | Required for name resolution.                                         |
| VLAN 50 – Student (if applicable) | NTP    | UDP     | 123       | Permit | Required for time sync / log integrity.                               |

## Section 3 – Management Plane Protection

| Source VLAN                            | Destination            | Access Type | Permit / Deny | Justification                                                                  |
|----------------------------------------|------------------------|-------------|---------------|--------------------------------------------------------------------------------|
| VLAN 1000 – Management                 | VLAN 1000 – Management | SSH         | Permit        | Only management network is authorized to administer devices via SSH.           |
| VLAN 30/40/50/200 (All non-Mgmt VLANs) | VLAN 1000 – Management | Any         | Deny          | Prevent client/service VLANs from reaching management plane (least privilege). |

## Section 4 – Inter-Client VLAN Restrictions

| Source VLAN        | Destination VLAN                | Permit / Deny | Justification                                                                                                    |
|--------------------|---------------------------------|---------------|------------------------------------------------------------------------------------------------------------------|
| VLAN 30 – Research | VLAN 40 – Faculty               | Deny          | Enforce segmentation / block lateral movement.                                                                   |
| VLAN 40 – Faculty  | VLAN 30 – Research              | Deny          | Enforce segmentation / block lateral movement.                                                                   |
| VLAN 30 – Research | VLAN 30 – Research (East-West?) | Permit        | East-West inside same VLAN is L2; not enforceable with SVI ACLs (would require L2 controls like PVLAN/microseg). |

**AAA & User Credential Table**

| <b>AAA &amp; User Credential Account Type</b> | <b>Device / Service</b> | <b>Username</b> | <b>Privilege Level / Role</b> | <b>Password</b> |
|-----------------------------------------------|-------------------------|-----------------|-------------------------------|-----------------|
| <b>TACACS+</b>                                | Network Devices         | netadmin        | 15                            | samliz29@@@     |
| <b>TACACS+</b>                                | Network Devices         | helpdesk        | 5                             | samliz29@@@     |
| <b>Local Fallback</b>                         | Switch/Router           | breakglass      | 1                             | samliz29@@@     |
| <b>RADIUS</b>                                 | Web01 Authentication    | webuser1        |                               | samliz29@@@     |
| <b>RADIUS</b>                                 | Web01 Authentication    | webuser2        |                               | samliz29@@@     |

Note: Same password was used, because it is a simulation environment and for easy login while working to, but at the implementation, same password can not be used, because different using passwords is very crucial for securing the role levels in the network